

The Sign of Beam Polarization

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The sign of beam polarization is obtained in each Moller measurement described in the tables for raw results. Also, the sign is quoted in the summary files for various experiments. This sign is defined within a certain convention. Unfortunately, this convention was not always the same for the period 1998-2000. This document describes how to obtain the polarization sign from the raw results.

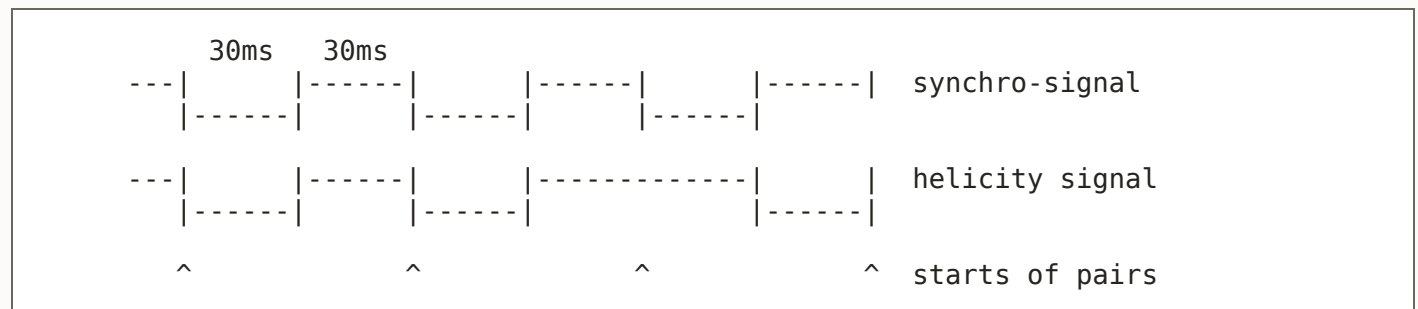
Helicity signals and beam polarization

The beam polarization at the injector is flipped at a certain frequency and with a certain pattern. This is achieved by flipping the circular polarization of the laser light with the Poekels cell. The pattern consists of pairs - the combinations of two periods of the same duration (30ms typically) and with opposite polarizations. Two modes can be used:

- *Toggle Mode*: each pair starts with the same polarity as the previous pair;
- *Random Mode*: each pair starts with a polarity chosen by a set of pseudo-random numbers.

Two signals arrive from the injector to the hall counting house. We call them here:

- *Synchro-signal*: a meander-type wave, alternating the logical levels 0 and 1. Each level pulse has a fixed duration - 30ms for most of the running time since 1998, but sometimes it was 1s.
- *Helicity signal*: this signal is tied to the state of the Poekels cell. The level can change only synchronously with the change in the *synchro-signal*.



Additionally to the fast spin flipping by the Poekels cell, the laser light polarization can be flipped by inserting a $1/2$ wave-length plate. This is done, typically, a few times per month, depending on the experimental requirements.

On the way from the injector to the hall the beam polarization may change its sign due to spin precession. The precession angle is well predictable.

The *helicity signal* should define the beam polarization sign at the injector unambiguously, for a given state of the $1/2$ -wave plate. However, a big change in the injector hardware or setting may alter the correlation.

Beam polarization in the hall

The only way to know for sure the beam polarization sign in a hall is to measure it in the hall using Moller/Compton polarimeters or some other spin-sensitive process like ep elastic scattering with the FPP (focal plane polarimeter) technique or ep scattering with a polarized target. The sign is measured within a certain *helicity convention*, namely the sign measured is assigned to a given phase of the *helicity signal*, say

to its logical 1. If several setups are involved, say the Moller polarimeter and the spectrometer, involving different hardware and software, one has to compare the *helicity conventions* for both setups.

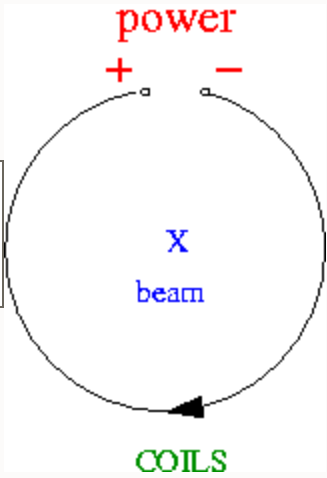
Important: if the experiment is relying on the polarization sign provided by the Moller polarimeter, the *helicity conventions* should be checked very thoroughly. For example, one should provide a certain helicity-asymmetric signal to the BCM lines of both setups and compare the net results of the asymmetry measurements.

For HAPPEX-1 run of 1998 the following procedure was used. The Moller raw signal was provided to the HAPPEX DAQ and data analysis stream, along with the information on the field direction. The beam polarization sign was determined within the HAPPEX *helicity convention*.

Hall A Moller polarimeter

Let us consider the longitudinal polarization only. The Moller cross-section is proportional to a term as follows: $(1-A \cdot P_b \cdot P_t)$, where A is about 0.7 while P_b and P_t are the beam and the target polarizations. Therefore, for parallel spins the counting rate is lower than for anti-parallel. The target polarization is defined by the direction of the magnetic field magnetizing the target. The field is provided by Helmholtz coils. At the positive voltage on the power supply the magnetic field $P_z < 0$. The current direction is given by a program and is stored in the Moller logfile for the given run. The connection hold from 1998 provides the field direction and therefore the polarization sign of the target as follows:

Current	Bz coils	Pz target
+	-	+



Therefore, if the Moller asymmetry measured for +9.0A in the coils is negative, the beam polarization in the given *helicity convention* is positive.

Interpretation of the Moller polarimetry results in Hall A

The polarization sign logged belongs to the following logical levels of the NIM *helicity signal*:

From	To	Logical level
summer 1998 -	30 Nov 2000	0
01 Dec 2000 -		1